

ILLINOIS TECH | College of Computing

ITMM 485 / 585
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Legal and Ethical Issues in
 Information Technology



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Ethical Concepts and Theories

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Ch8: Intellectual Property Issues
P3: Patents, Trademarks
and Trade Secrets



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Learning Objectives:

Upon completion of this lesson the students should be able to:

- Explain and discuss the philosophical and legal arguments used to protect intellectual property
- Describe the differences and similarities between the Free Software Foundation & the Open Source Initiative
- Explain how Creative Commons licensing of IP works
- Discuss the balance between public good and profit for IP rights holders

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Module Objectives:

Upon completion of this lesson the students should be able to:

- Define intellectual property (IP) and related terms including patents, trademarks & trade secrets

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Patent Protections

- A patent is a form of legal protection given to individuals who create an invention or process.
- Unlike copyrights, patents offer a 20-year exclusive monopoly over an expression or implementation of a protected work.
- Current U.S. patent law is based on the Patent Act of 1952, amended in 1995.

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Patents (Continued)

- Patent protection can be applied to inventions and discoveries that include utilitarian or functional devices such as machines, "articles of manufacture," or "compositions of matter."
- Patents are granted to inventions and discoveries that satisfy three conditions:
 - 1) *usefulness*,
 - 2) *novelty*,
 - 3) *non-obviousness*.

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Patents (Continued)

- Computer hardware inventions clearly satisfied the requirements of patent law.
- Computer software did not (initially).
- Beginning with *Gotshalk v. Benson* (1972), the U.S. Patent Office and the courts established a strong opposition to patents.
- Benson applied for a patent for an algorithm he developed that translated the representation of numbers from base 10 to base 2.

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Gotshalk vs. Benson

- Benson's algorithm is an important feature of all programs.
- If he had been granted a patent for his algorithm, Benson would have controlled almost every computer in use for 12 years.
- The patent was denied to Benson on the basis of a policy that bars patents for mere mathematical formulas or abstract processes that can be performed by a series of "mental steps" with the aid of pencil and paper.

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The *Diamond v. Diehr* Case

- The *Diamond v. Diehr* case is considered a landmark ruling in the dispute over patenting software.
- The outcome of this court case, based on 5-4 decision, resulted in the first patent awarded for a computer program.
- In this instance, the computer program assisted in a process of converting rubber into tires.
- Critics argued that Diehr had only a new computer program, since all of the parts of the machine used in the conversion process consisted of traditional technology except for the computer program.

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Diamond v. Diehr (Continued)

- Although the Court ruled in favor of Diehr, the justices, in their decision, continued to affirm the view that computer algorithms themselves are not patentable.
- The Court pointed out that the patent awarded to Diehr was not for the computer program but for the rubber tire transformation process as a whole.

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Proliferation of Patents for Computer Software

- Some worry that patent protection has gone too far.
- The U.S. Patent and Trademark Office (PTO) issues about 20,000 new software patents every year.
- Aharonian (2001) notes that between 1993 and 1999, the number of patents issued increased by tenfold.
- Between 1979 and 1999, more than 700,000 patents had been issued for electronics inventions.

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Trademarks

- A *trademark* is a word, name phrase, or symbol that identifies a product or service.
- The Lanham Act, also referred to as the Trademark Act of 1946, was passed to provide protection for registered trademarks.
- The Act intends to ensure that the quality associated with a certain logo or symbol used by a business actually represents the quality that consumers expect (e.g., the BMW label).

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Trademarks (Continued)

- To qualify for a trademark, the "mark" or name is supposed to be *distinctive*.
- However, Deborah Halbert (1999) notes that a (not so distinctive) trademark for "uh-huh" was granted to Pepsi.
- Also, consider that a major movie theatre in the U.S. has trademarked the expression "Silence is Golden."

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AOL v. AT&T

- America OnLine (AOL) tried to register a number of symbols as official trademarks.
- For example, it applied for trademarks of its expression "You've Got Mail," "Buddy List," and "IM" (for Instant Messenger).
- If AOL was allowed to register them, other providers who used these expressions would be infringing on AOL's registered trademarks.
- AT&T challenged AOL.
- The court ruled that the expressions were not unique to AOL.

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Trade Secrets

- A trade secret is defined as information used in the operation of a business or other enterprise that is sufficiently valuable and secret to afford an actual or potential economic advantage over others.
- Trade secrets can be used to protect:
 - formulas (such as the one used by Coca-Cola);
 - blueprints for future projects;
 - chemical compounds;
 - process of manufacturing.

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Jurisdictional Issues Involving Copyright Laws

- Intellectual property (IP) laws enacted in the U.S. have implications that are global.
- Some international treaties for IP have been signed, such as TRIPS (Trade Relationship Aspects of Intellectual Property Standards).
- The TRIPS agreement implemented requirements from the Berne Convention for the Protection of Literary and Artistic Works.
- TRIPS is recognized by signatories to the World Intellectual Property Organization (WIPO).

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State Laws in the U.S. Affecting Goods and Services

- Different states in the U.S. have different laws affecting the sale-of-goods and contracts involving goods and services.
- One controversial law (at the state level) affecting software is the Uniform Computer and Information Transactions ACT (UCITA).
- UCITA is designed to govern contracts for the development, sale, licensing, maintenance, and support of computer software.

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UCITA

- UCITA also extends to all shrink-wrap and "click-wrap" agreements.
- Thus far, UCITA has been enacted into law in the states of Virginia and Maryland
- But UCITA has implications for all states because contracts span multiple states and thus potentially involve Virginia and Maryland law in the process.

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